JOONHYUK SEO

EDUCATION

Hanyang University, Seoul, Republic of Korea
Master of Science, Department of Artificial Intelligence, Aug 2025 (Expected)
Cumulative GPA: 3.83/4.0
Relevant Coursework: Optimization Theory (A), Advanced Machine Learning Theory (A), Introduction to Random Processes (A)

Hanyang University, Ansan-si, Gyeonggi-do, Republic of Korea
Bachelor of Science, Majored in Media Technology, March 2017-Feb 2023
- ✧ Military Service, September 2018 – May 2020
  Academic hiatus due to military service

RESEARCH EXPERIENCE

Electromagnetics & Intelligent Design Lab
Hanyang University, Seoul, Republic of Korea
Graduate Research Student, March 2023 – Present

Publications

[P1] Deep-learning-driven end-to-end metalens imaging [Advanced Photonics]
Advanced Photonics (IF: 20.6) – Top-tier journal in Optics.
†Joonhyuk Seo, †J Jo, †J Kim, †J Kang, C Kang, S Moon, E Lee, J Hong, J Rho, H Chung

[P2] Wave Interpolation Neural Operator: Interpolated Prediction of Electric Fields Across Untrained Wavelengths [NIPS workshop]
NIPS Workshop 2024
Generalizing Deep Surrogate Solvers for Broadband Electromagnetic Field Prediction at Unseen Wavelengths [Preprint]
(Under Review),
†Joonhyuk Seo, †C Kang, D Seo, H Chung

[P3] Adjoint Method-based Fourier Neural Operator Surrogate Solver for Wavefront Shaping in Tunable Metasurfaces
iScience (IF: 6.1)
†C Kang, †Joonhyuk Seo, Ikbeom Jang, H Chung

[P4] High-Speed Multiwavelength Adjoint Optimization with Surrogate Solver
Pacific Rim Conference on Lasers and Electro-Optcis (**CLEO-PR**)
†Joonhyuk Seo, †C Kang, D Seo, H Chung

Research: Application of deep learning to various tasks in photonics

- ✧ Developed an image restoration model to alleviate severe and complex degradations, which is a highly ill-posed inverse problem[P1].
  1) Employed an auxiliary generative model, a generative adversarial network (GAN), in order to reflect the data distribution of latent clean images.
  2) Elevated the PSNR by 7.37 dB and SSIM by 22.8%p compared to the original metalens images.

- ✧ A novel electromagnetic surrogate solver that enables the interpolated prediction of unseen wavelengths from simulation observations of discrete wavelength[P2].
  1) Created an electromagnetic simulation dataset through a Finite-Difference Frequency-Domain simulator (Ceviche).
  2) Proposed a conditioning method for strong regularization and a parameter-efficient neural operator architecture.
  3) A 74% reduction in parameters and 80.5% improvements in prediction accuracy for untrained wavelengths, and 13.2% improvements for trained wavelengths.

Vision and Machine Learning Lab
Hanyang University, Ansan-si, Republic of Korea
Undergraduate Research Student, July 2021 – March 2022

- ✧ Supported main researchers' works by reading and reproducing the effective GAN-based models and employing various datasets (LSUN datasets, Pokemon, etc.) and metrics (Inception Score, Fréchet Inception Distance) for training and evaluating GAN-based models.
- ✧ Studied Variational Autoencoders (VAE), including variational inference theory, ß-VAE, and Vector Quantization VAE to understand how VAE works and VAE's representation learning capabilities. (*GitHub)*
- ✧ Studied denoising score matching methods and diffusion probabilistic models (DPM), including a score-based generative model, Variance-Exploding and Variance-Preserving SDE, and solving inverse problems with score-based generative models, to comprehend how these models work and how to employ score-based generative models and DPMs to solve ill-posed inverse problems. (*GitHub)*

PATENT

Deep Learning Image Restoration Method for Ultra-Compact Imaging Systems
Korean Patent
Inventors: Joonhyuk Seo, J Jo, D Seo, C Kang, H Chung

HONORS AND AWARDS

[A1] CES 2025 Innovation Award: 1) XR and 2) Mobile Device [News]
Subject: Super-compact Metalens Imaging via AI-driven Hardware-Software
January 2025

[A2] Corning and Hanyang University: Corning AI Challenge (2th) [Homepage]
Subject: Estimating adjoint gradients using AI for optical structure optimization.
December 2023

[A3] Ministry of Science and ICT of South Korea: AI Grand Challenge, Policy Assistance AI
(President's Award from Institute for Information & communication Technology Planning & evaluation) [Press]
Subject: Developing an AI for the interpretation of governmental documents using Natural Language Processing and Computer Vision methods.
July 2023

[A4] The Journal of Korean Institute of Electromagnetic Engineering and Science:
Outstanding Paper at the 2023 Fall Academic Conference [Hompage]
Subject: Deep learning image restoration models for metalens imaging
Novemeber 2023

[A5] Hanyang University : HY-Innovation Award (Grand Prize, 1th) [Press]
Subject: Devising co-optimization between a deep learning image reconstruction model
and optics structures for metalens that causes severe image degradation
July 2023

[A6] Hanyang University: AI Contest to Predict Credit Card User Delinquency (Excellence Prize)
December 2021

[A7] Korea Gas Corporation : Gas Demand Estimation Competition (Final round 4th / 488)
December 2021